

SHOWMICK DAS

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EDUCATION

Georgia Institute of Technology

Atlanta, GA

Bachelor of Science in Computer Science, GPA: 4.00

Expected Graduation, May 2027

- **Relevant Coursework:** Artificial Intelligence, Computer Organization & Programming, Database Systems, Data Structures & Algorithms, Discrete Mathematics, Objects & Design, Object-Oriented Programming, Computing in Python, Linear Algebra, Calculus I-III

EXPERIENCE

Georgia Institute of Technology – Artificial Intelligence Lab

Dec 2024 – Present

Software Engineer

Atlanta, GA

- Designed **5+ REST endpoints** with reverse proxy support for model loading, dataset uploads, and fine-tuning on an **NVIDIA DGX server**.
- Built inference dashboard with **React.js & Tailwind CSS** achieving **60ms latency** for **Large Pre-Trained Time-Series Models (LPTMs)**.
- Evaluated and benchmarked **LPTMs** against **5 Foundational Models** including **Chronos** and **TimesFM** using **1.6M+ data points**.
- Contributed to research published at **NeurIPS 2024**, one of the **most prestigious AI conferences** in the US.

University of Georgia – Dept. of Epidemiology & Biostatistics

May 2025 – Aug 2025

Software Engineering Intern

Athens, GA

- Trained **XGBoost peak classification model** achieving **98.4% accuracy** for automated **heartbeat signal processing**.
- Improved accuracy on **76.8%** of **biomedical predictions** while reducing training time by **16%** using **NVIDIA T4 GPU acceleration**.
- Architected **1D CNN classifier** for **signal quality assessment** achieving **0.9917 F1 score** via **Bayesian Optimization**.
- Integrated **CNN classifier** as **preprocessing head** reducing **BiLSTM model's RMSE** by **65%** and eliminating training instabilities.

The Apache Software Foundation

May 2024 – Aug 2024

Software Engineering Intern

Wilmington, DE

- Built **FastAPI-based MCP server** serving **5K+ concurrent researchers** with **100% uptime** and **OAuth2 authentication**.
- Integrated **LangChain AI Agent** powered by **open-source Qwen3 LLM** into Apache's **\$5M NSF-funded** research platform, **Cybershuttle**.
- Engineered **20+ REST endpoints** transforming complex **Apache Airavata API workflows** into intuitive conversational interfaces.
- Enabled natural language queries across **100K+ datasets, models, and notebooks** at **GPU-accelerated research clusters**.

LEADERSHIP

Perplexity AI

Aug 2025 – Present

Campus Ambassador

San Francisco, CA

- Driving **500+ sign-ups** for **Comet**, Perplexity's new AI-powered browser, through campus outreach and student organizations.

Artificial Intelligence at Georgia Tech

Aug 2025 – Present

Executive Board of Finance & Sponsorships

Atlanta, GA

- Managing **\$50K+** funding from **Google Cloud, Microsoft, & Anthropic** for **AI ATL**, Atlanta's largest AI hackathon with **600+ hackers**.

Stanford University

Apr 2025 – May 2025

Teaching Assistant

Stanford, CA

- Led weekly **Python** coding sessions on **data structures** and **object-oriented programming**, achieving **16.7%** improved attendance.

Propulsive Landers at Georgia Tech

Aug 2024 – Dec 2024

Executive Board of Website Development & Marketing

Atlanta, GA

- Built marketing website using **React.js & Tailwind CSS**, securing **4 new sponsors**, **\$5K** in funding, and **28.7%** increase in engagement.

PROJECTS

JurassIQ | React.js, Node.js, TensorFlow.js, Scikit-learn, OpenCV, Folium, FastAPI, OpenAI API

- Developed an **AI archaeology engine** using a fine-tuned **MobileNetV2** to predict dig sites and fossil value from **100K+ images**.
- Integrated an **AI Agent** with a **React.js & Tailwind CSS dashboard** to generate excavation plans covering **cost, logistics, and duration**.
- Visualized **240K+ geospatial data points** as interactive heatmaps using **Folium** to identify fossil distribution patterns.
- Engineered and deployed the full-stack application in **36 hours** for **Hacklytics 2025** at **Georgia Institute of Technology**.

Fine-tuning Code Llama For FastAPI Code Generation | Python, PyTorch, Hugging Face, GitHub API, OpenAI API

- Engineered an automated data pipeline to mine **120+ FastAPI repositories**, extracting code patterns via **AST parsing**.
- Fine-tuned **Code Llama** using **4-bit quantization** and **LoRA adapters**, reducing GPU memory usage by **75%** and training time by **50%**.
- Validated a **13.7%** improvement in code generation accuracy using a custom **LLM-based evaluation framework**.
- Authored the **most downloaded open-source Code Llama adapter** on the **Hugging Face Hub** for FastAPI code generation.

Implementing Attention Is All You Need | Python, PyTorch, pytest, Hugging Face, Weights & Biases

- Implemented the **Transformer architecture from scratch** in PyTorch, achieving a **14.8 BLEU score** on English-Italian translation.
- Engineered a resource-efficient training loop achieving **35% of benchmark performance** on a **single A100 GPU**.
- Built a data pipeline to process **1.25M+ sentence pairs** and integrated **Weights & Biases** for experiment tracking.

TECHNICAL SKILLS

Languages: Python, C++, Java, SQL, Bash, JavaScript, TypeScript, Kotlin, HTML/CSS, C, C#, Assembly

Technologies: React.js, Tailwind CSS, Node.js, FastAPI, Flask, Spring Boot, Git, Docker, Kubernetes, PostgreSQL, MongoDB, Firebase, AWS,

Microsoft Azure, Google Cloud Platform, Android Studio, PyTorch, Scikit-learn, Hugging Face, LangChain, CUDA, Pandas, NumPy, Kafka, Spark

Concepts: Machine Learning, Large Language Models, Time Series Forecasting, Data Science & Analysis, Distributed Systems, Database Design, RESTful APIs, Cloud Computing, Computer Architecture, CI/CD Pipelines, Unit Testing, A/B Testing, Website & Mobile Development